

Alternating Current and Electromagnetic Waves

Oscillating currents and the waves they launch: AC circuits, resonance, Maxwell's equations and light itself.

By **Pratik Bhanuse**, PhD Physics candidate, Western University · pratikbhanuse.github.io

Mains electricity is **alternating current** (AC): its voltage swings sinusoidally, 50 or 60 times a second. In AC circuits, inductors and capacitors no longer just store energy. They impede current in frequency-dependent ways, and circuits that contain both can **resonate**, which is the basis of radio tuning, filters and wireless power. The second half of the chapter makes one of the great leaps in physics. Maxwell completed the laws of electromagnetism, and from them predicted **electromagnetic waves** travelling at exactly the measured speed of light. Light, radio, X-rays and the RF pulses of an MRI scanner are all the same phenomenon.

YOU WILL BE ABLE TO

- Use rms values, and the phase relations between voltage and current for R, L and C.
- Calculate inductive and capacitive reactance, impedance and phase angle for series RLC circuits.
- Analyse resonance, quality factor and bandwidth, and apply them to tuning and filters.
- Calculate average power and power factor, and explain power-factor correction.
- Describe LC oscillations as an electrical analogue of SHM.
- State Maxwell's equations, including the displacement current.
- Describe electromagnetic waves: their speed, the relation $\mathbf{E} = c\mathbf{B}$, intensity, radiation pressure and the spectrum.

THINK FIRST

In an MRI scanner the radio-frequency pulses at 3 T are at 128 MHz, which has a wavelength of 2.3 m in air. Yet radiologists find the images at 3 T can show bright and dark patches across the abdomen that are absent at 1.5 T. What do wavelengths have to do with it?

Inside the body, radio waves travel much more slowly than in air, because tissue has a high dielectric constant ($\kappa \approx 50\text{--}80$). The wavelength shrinks by roughly $\sqrt{\kappa}$, to about 30 cm at 128 MHz. That is comparable to the size of the torso, so the RF field forms partial **standing waves** inside the body, stronger in some places and weaker in others. This produces uneven tipping of the protons and patchy brightness. At 1.5 T (64 MHz) the wavelength in tissue is about twice as long, and the effect is much weaker. Engineers counter it with “parallel transmit” coils and high-permittivity pads.

§21.1

Alternating voltages and currents

An AC source produces $v(t) = V_0 \sin \omega t$, where $\omega = 2\pi f$. In Europe, India and much of the world, $f = 50$ Hz. In North America, $f = 60$ Hz.

Root-mean-square values. The heating effect of a current depends on i^2 , so we characterise AC by the steady (DC) value that would give the same average power. The average of $\sin^2$ over a cycle is $\frac{1}{2}$, so

$$I_{\text{rms}} = \sqrt{\langle i^2 \rangle} = \frac{I_0}{\sqrt{2}}, \quad V_{\text{rms}} = \frac{V_0}{\sqrt{2}}.$$

“230 V mains” means $V_{\text{rms}} = 230$ V, with peaks of ± 325 V. Meters read rms values, and power formulas use them: $\bar{P} = I_{\text{rms}}^2 R$.

WORKED EXAMPLE 21.1**Peak values in mains supply**

A 2.0 kW kettle runs on 230 V rms mains. Find the rms and peak currents and the peak voltage.

Solution

$$I_{\text{rms}} = \frac{2000}{230} = 8.7 \text{ A}, \quad I_0 = \sqrt{2}(8.7) = 12.3 \text{ A}, \quad V_0 = \sqrt{2}(230) = 325 \text{ V}.$$

Insulation must withstand the *peak* voltage, not just the rms value.

§21.2

Resistors, inductors and capacitors in AC circuits

Apply $i = I_0 \sin \omega t$ to each element in turn.

- **Resistor:** $v = iR$. The voltage is **in phase** with the current.
- **Inductor:** $v = L di/dt = \omega L I_0 \cos \omega t$. The voltage **leads** the current by 90° . Its amplitude is $V_0 = I_0 X_L$, with inductive reactance

$$X_L = \omega L.$$

- **Capacitor:** $v = q/C$, with $q = \int i dt = -\frac{I_0}{\omega} \cos \omega t$. The voltage **lags** the current by 90° . Its amplitude is $V_0 = I_0 X_C$, with capacitive reactance

$$X_C = \frac{1}{\omega C}.$$

Reactance is measured in ohms. An inductor impedes **high** frequencies and lets DC through easily. A capacitor blocks DC and passes high frequencies. A pure inductor or capacitor absorbs no **average** power: energy flows in during one quarter-cycle and back out during the next.

Mnemonic: ELI the ICE man. In an inductor (L), the EMF E leads the current I . In a capacitor (C), the current I leads the EMF E .

WORKED EXAMPLE 21.2

Reactance depends on frequency

Find the reactances of a 0.10 H inductor and a 10 μF capacitor at 50 Hz and at 5.0 kHz.

Solution

	50 HZ	5.0 KHZ
$X_L = 2\pi fL$	31 Ω	3.1 k Ω

	50 HZ	5.0 KHZ
$X_C = 1/(2\pi fC)$	320 Ω	3.2 Ω

A 100-fold increase in frequency multiplies X_L by 100 and divides X_C by 100. This frequency dependence is what makes filters possible.

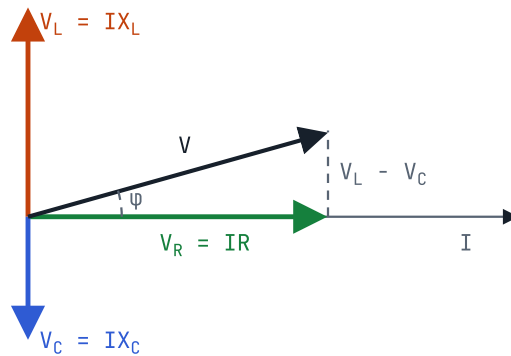
§21.3

The series RLC circuit

With R , L and C in series, the same current $i = I_0 \sin \omega t$ flows through each, but their voltages are out of phase with one another. Represent each voltage as a rotating vector, a **phasor**:

- V_R is along the current phasor;
- V_L is 90° ahead of it;
- V_C is 90° behind it.

The phasors add like vectors. V_L and V_C point in opposite directions, so they partly cancel:



Phasor diagram for a series RLC circuit (here with $X_L > X_C$). The source voltage V is the vector sum of V_R and $V_L - V_C$, and leads the current by the phase angle ϕ .

From the right triangle:

$$V = I\sqrt{R^2 + (X_L - X_C)^2} = IZ, \quad Z = \sqrt{R^2 + (X_L - X_C)^2}, \quad \tan \phi = \frac{X_L - X_C}{R}.$$

Z is the impedance. If $X_L > X_C$, the circuit is inductive and the voltage leads the current ($\phi > 0$). If $X_C > X_L$, it is capacitive and the current leads ($\phi < 0$).

Resonance

The impedance is smallest, and the current largest, when $X_L = X_C$:

$$\omega_0 = \frac{1}{\sqrt{LC}}, \quad f_0 = \frac{1}{2\pi\sqrt{LC}}.$$

At resonance, $Z = R$, the voltage and current are in phase, and V_L and V_C cancel exactly. Each of them can still be *much larger* than the source voltage. The **quality factor**

$$Q = \frac{\omega_0 L}{R} = \frac{1}{R} \sqrt{\frac{L}{C}}$$

measures the sharpness of the resonance. The bandwidth between the half-power points is $\Delta\omega = \omega_0/Q = R/L$. This is the electrical counterpart of the driven oscillator in Chapter 9: L plays the role of mass, $1/C$ of stiffness, and R of damping.

WORKED EXAMPLE 21.3

A series RLC circuit on the mains

A series circuit with $R = 100 \, \Omega$, $L = 0.50 \, \text{H}$ and $C = 10 \, \mu\text{F}$ is connected to 230 V, 50 Hz mains. Find the impedance, the current, the phase angle, the power factor and the average power. What is the resonant frequency?

Solution

Reactances:

$$X_L = 2\pi(50)(0.50) = 157 \, \Omega, \quad X_C = \frac{1}{2\pi(50)(10 \times 10^{-6})} = 318 \, \Omega.$$

Impedance:

$$Z = \sqrt{100^2 + (157 - 318)^2} = \sqrt{10\,000 + 25\,990} = 190 \, \Omega.$$

Current and phase:

$$I_{\text{rms}} = \frac{230}{190} = 1.21 \, \text{A}, \quad \tan \phi = \frac{-161}{100} \Rightarrow \phi = -58^\circ.$$

The circuit is capacitive, so the current leads the voltage.

Power: the power factor is $\cos \phi = 0.53$, and

$$\bar{P} = I_{\text{rms}}^2 R = (1.21)^2 (100) = 147 \text{ W.}$$

Check: $V_{\text{rms}} I_{\text{rms}} \cos \phi = 230(1.21)(0.53) = 147 \text{ W. } \checkmark$

Resonance:

$$f_0 = \frac{1}{2\pi \sqrt{(0.50)(10 \times 10^{-6})}} = 71 \text{ Hz.}$$

WORKED EXAMPLE 21.4

Tuning a radio

An FM receiver's tuning circuit has $L = 2.0 \mu\text{H}$. What capacitance tunes it to a station at 100 MHz?

Solution

$$C = \frac{1}{\omega_0^2 L} = \frac{1}{(2\pi \times 10^8)^2 (2.0 \times 10^{-6})} = 1.3 \times 10^{-12} \text{ F} = 1.3 \text{ pF.}$$

Turning a variable capacitor, or adjusting the voltage on a varactor diode, shifts f_0 across the band. Only the station at resonance drives a large current, so the others are rejected.

§21.4

Power in AC circuits

The instantaneous power is $p = vi$. Averaged over a cycle, with a phase difference ϕ between v and i :

$$\bar{P} = V_{\text{rms}} I_{\text{rms}} \cos \phi.$$

The **power factor** $\cos \phi$ is 1 for a pure resistor and 0 for a pure inductor or capacitor. Motors and fluorescent-lamp ballasts are inductive, with power factors around 0.7–0.8. They draw more current than their real power requires. That extra current heats the supply cables and transformers without doing useful work, so utilities penalise low power factors. Adding capacitors in parallel cancels the inductive reactance: this is **power-factor correction** (Problem P21.8).

§21.5

LC oscillations

A charged capacitor connected to an inductor (with no resistance) oscillates. The charge flows out through the inductor, overshoots, charges the capacitor the other way, and so on. The loop equation is

$$L \frac{d^2 q}{dt^2} + \frac{q}{C} = 0 \quad \Rightarrow \quad q = Q_0 \cos \omega_0 t, \quad \omega_0 = \frac{1}{\sqrt{LC}}.$$

This is exactly the SHM equation of Chapter 9. Energy sloshes between the capacitor's electric field, $\frac{q^2}{2C}$ (like potential energy), and the inductor's magnetic field, $\frac{1}{2} Li^2$ (like kinetic energy). The total stays constant: $\frac{Q_0^2}{2C} = \frac{1}{2} LI_0^2$. With resistance included, the oscillations decay, just like a damped oscillator.

WORKED EXAMPLE 21.5

An LC oscillator

A 1.0 μF capacitor charged to 10 V is connected across a 10 mH inductor. Find the oscillation frequency and the maximum current.

Solution

$$f = \frac{1}{2\pi\sqrt{(10^{-2})(10^{-6})}} = \frac{1}{2\pi(10^{-4})} = 1.6 \text{ kHz}.$$

Energy conservation gives $\frac{1}{2} CV_0^2 = \frac{1}{2} LI_0^2$, so

$$I_0 = V_0 \sqrt{\frac{C}{L}} = 10 \sqrt{\frac{10^{-6}}{10^{-2}}} = 0.10 \text{ A}.$$

IN PRACTICE

Filters in medical electronics. An ECG signal is about 1 mV, with useful frequencies from about 0.05 Hz to 150 Hz. It is buried in interference: a slow baseline drift from breathing and electrode movement, high-frequency muscle noise, and strong mains “hum” at 50 or 60 Hz picked up from the room. Simple RC filters shape the signal. A **high-pass** filter with cut-off $f_c = 1/2\pi RC \approx 0.05$ Hz removes the drift, a **low-pass** filter at about 150 Hz removes the high-frequency noise, and a narrow **notch** filter (a resonant circuit, usually implemented digitally) removes the mains hum (Problem P21.18). Similar filters condition EEG, EMG and pulse-oximeter signals, and the anti-aliasing stage of every digital audio and imaging system.

§21.6

Maxwell's equations

By the 1860s, the laws of electricity and magnetism could be summarised in four equations. James Clerk Maxwell noticed that Ampère's law was **inconsistent** for charging capacitors. A current flows *into* the capacitor's plates, but no current crosses the gap between them, so the result of Ampère's law depended on which surface was used to count the enclosed current. Maxwell resolved this by adding a **displacement current**, $I_d = \epsilon_0 d\Phi_E/dt$. A *changing electric field* produces a magnetic field, just as a changing magnetic field produces an electric field. Between the plates, I_d exactly equals the conduction current in the wires (Problem P21.14).

KEY IDEA

Maxwell's equations (integral form, in vacuum):

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{enc}}}{\epsilon_0} \quad (\text{Gauss: charges make } \vec{E})$$

$$\oint \vec{B} \cdot d\vec{A} = 0 \quad (\text{no magnetic monopoles})$$

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt} \quad (\text{Faraday: changing } \vec{B} \text{ makes } \vec{E})$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}} + \mu_0 \epsilon_0 \frac{d\Phi_E}{dt} \quad (\text{Ampère–Maxwell: currents and changing } \vec{E} \text{ make } \vec{B})$$

Together with the Lorentz force, $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$, they describe all classical electromagnetic phenomena.

§21.7

Electromagnetic waves

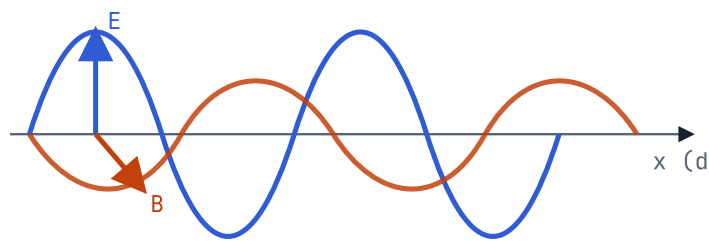
In empty space, with no charges or currents, the last two equations feed each other. A changing $\vec{E}$ makes a $\vec{B}$, which changes and makes an $\vec{E}$, and so on, so the disturbance propagates. Maxwell showed that both fields obey the wave equation (Chapter 11) with speed

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = \frac{1}{\sqrt{(4\pi \times 10^{-7})(8.854 \times 10^{-12})}} = 3.00 \times 10^8 \text{ m/s.}$$

This matched the measured speed of light. Maxwell concluded (1865) that **light is an electromagnetic wave**. Heinrich Hertz generated and detected radio waves in 1887, confirming the prediction.

Properties of a plane EM wave, for example $\vec{E} = E_0 \sin(kx - \omega t) \hat{j}$ and $\vec{B} = B_0 \sin(kx - \omega t) \hat{k}$:

- $\vec{E}$ and $\vec{B}$ are perpendicular to each other and to the direction of travel. The wave is **transverse**, and it travels in the direction of $\vec{E} \times \vec{B}$.
- $\vec{E}$ and $\vec{B}$ oscillate in phase, with $E = cB$ at every instant.
- $c = f\lambda$ for every frequency. In a material with dielectric constant κ (and $\mu_r \approx 1$), the speed is $v = c/\sqrt{\kappa} = c/n$, where n is the refractive index (Chapter 22).



$$E \perp B \perp x, E = cB, \text{ in pl}$$

A plane electromagnetic wave. E (blue) and B (orange, drawn in perspective) oscillate perpendicular to each other and to the direction of travel, in phase, with $E = cB$.

Energy, intensity and momentum

The wave carries energy in both fields, shared equally between them: $u = \frac{1}{2}\epsilon_0 E^2 + \frac{B^2}{2\mu_0} = \epsilon_0 E^2$ at each instant. The rate of energy flow per unit area is given by the **Poynting vector**,

$\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B}$. Averaged over time, this gives the **intensity**:

$$I = \frac{E_0 B_0}{2\mu_0} = \frac{1}{2} c \epsilon_0 E_0^2 = c \epsilon_0 E_{\text{rms}}^2.$$

EM waves also carry **momentum**. A wave absorbed by a surface exerts a **radiation pressure** $p = I/c$. If it is perfectly reflected, the pressure is $2I/c$.

WORKED EXAMPLE 21.6

The fields in sunlight

Sunlight at the Earth's surface has an intensity of about 1000 W/m^2 . Find the rms electric and magnetic fields, and the radiation pressure on a black surface.

Solution

$$E_{\text{rms}} = \sqrt{\frac{I}{c\epsilon_0}} = \sqrt{\frac{1000}{(3.0 \times 10^8)(8.854 \times 10^{-12})}} = 610 \text{ V/m}, \quad B_{\text{rms}} = \frac{E_{\text{rms}}}{c} = 2.0 \times 10^{-6} \text{ T}$$

$$p = \frac{I}{c} = \frac{1000}{3.0 \times 10^8} = 3.3 \times 10^{-6} \text{ Pa}.$$

Evaluate. The pressure is tiny on Earth, but in space it steadily pushes on spacecraft and solar sails (Problem P21.16), and it helps shape comet tails.

The electromagnetic spectrum

All electromagnetic waves travel at c in vacuum. They differ only in frequency and wavelength, and therefore in how they interact with matter.

BAND	WAVELENGTH	TYPICAL SOURCES AND USES, INCLUDING MEDICAL
Radio	$> 1 \text{ m}$	Broadcasting; MRI RF coils (64–300 MHz)
Microwave	$1 \text{ mm} - 1 \text{ m}$	Wi-Fi, phones, radar, ovens (2.45 GHz); microwave

BAND	WAVELENGTH	TYPICAL SOURCES AND USES, INCLUDING MEDICAL
		ablation of tumours
Infrared	700 nm – 1 mm	Thermal radiation (Chapter 12); thermography , remote controls, fibre optics
Visible	400–700 nm	Vision, photosynthesis; endoscopy, pulse oximetry (red and IR)
Ultraviolet	10–400 nm	Sun tanning, vitamin D synthesis; UV sterilisation ; causes skin cancer
X-rays	0.01–10 nm	Radiography, CT ; crystallography
Gamma rays	< 0.01 nm	Nuclear decay; radiotherapy , PET (511 keV), sterilisation

From UV upwards, each photon carries enough energy to ionise atoms and break chemical bonds (Chapter 24). That is the dividing line between **ionising** and **non-ionising** radiation, and the main reason for the very different safety rules for X-rays and for radio waves.

CHAPTER SUMMARY

- $V_{\text{rms}} = V_0/\sqrt{2}$ and $I_{\text{rms}} = I_0/\sqrt{2}$. Mains: 230 V or 120 V rms.
- Resistor: v in phase with i . Inductor: v leads by 90° , $X_L = \omega L$. Capacitor: v lags by 90° , $X_C = 1/\omega C$.
- Series RLC: $Z = \sqrt{R^2 + (X_L - X_C)^2}$ and $\tan \phi = (X_L - X_C)/R$. Resonance at $\omega_0 = 1/\sqrt{LC}$, where $Z = R$. $Q = \omega_0 L/R$, bandwidth ω_0/Q .
- Average power $\bar{P} = V_{\text{rms}} I_{\text{rms}} \cos \phi$. Power factor $\cos \phi$.
- LC circuit: SHM with $\omega_0 = 1/\sqrt{LC}$, with energy exchanged between E and B fields.
- Maxwell: the displacement current $\epsilon_0 d\Phi_E/dt$ completes Ampère's law, and the four equations predict EM waves.
- EM waves: $c = 1/\sqrt{\mu_0 \epsilon_0}$, transverse, $E = cB$, $I = \frac{1}{2} c \epsilon_0 E_0^2$, radiation pressure I/c (absorbed) or $2I/c$ (reflected).

Practice problems

Full step-by-step solutions are in the separate solutions PDF.

LEVEL A – CONCEPT CHECK

P21.1

Why are AC voltages and currents usually quoted as rms values rather than peak values? What would be the peak voltage of a 120 V supply?

P21.2

Explain, using reactance, why a capacitor blocks steady DC but passes AC, and why its opposition to current falls as the frequency rises.

P21.3

At resonance in a series RLC circuit, what are the impedance and the phase angle? Explain how the voltages across the inductor and the capacitor can each be much larger than the source voltage.

P21.4

Why must electromagnetic waves in vacuum be transverse? What is the direction of travel of a wave whose $\vec{E}$ points along $+y$ and $\vec{B}$ along $+z$?

LEVEL B – STANDARD PROBLEMS**P21.5**

A 0.20 H inductor with negligible resistance is connected to 230 V, 50 Hz mains. Find its reactance, the rms current, and the average power drawn.

P21.6

A 50 μF capacitor is connected to 230 V, 50 Hz mains. Find its reactance and the rms current. How does the current change if the frequency is doubled?

P21.7

A series circuit with $R = 20\ \Omega$, $L = 0.10\ \text{H}$ and $C = 50\ \mu\text{F}$ is connected to a 120 V, 60 Hz supply. Find the impedance, the rms current, the phase angle, the power factor, the average power, and the resonant frequency.

P21.8

Power-factor correction. A motor draws 10 A rms from a 230 V, 50 Hz supply at a power factor of 0.70 (lagging). (a) Find the real power and the reactive power. (b) What capacitance, connected in parallel with the motor, would raise the power factor to 1.0? (c) What is the supply current after correction?

P21.9

An LC circuit has $L = 25\ \text{mH}$ and $C = 4.0\ \mu\text{F}$. The capacitor is initially charged to 50 V. Find the oscillation frequency, the maximum charge and the maximum current.

P21.10

A receiver's tuning inductor is 0.30 μH . What range of capacitance is needed to tune across the FM band, from 88 MHz to 108 MHz?

P21.11

A 5.0 mW laser pointer produces a beam 1.0 mm in diameter. Find its intensity (compare it with sunlight, about 1000 W/m²), and the peak electric field in the beam. Why can even a low-power laser damage the retina?

P21.12

A microwave oven operates at 2.45 GHz. (a) Find the wavelength. (b) Standing waves form inside the oven. How far apart are the “hot spots”? How could you use a bar of chocolate and a ruler to measure the speed of light? (c) Why does the metal mesh in the door, with holes about 2 mm across, block the microwaves while letting visible light through?

LEVEL C – CHALLENGE PROBLEMS**P21.13**

(a) Show that the rms current in a series RLC circuit is $I = V / \sqrt{R^2 + (\omega L - 1/\omega C)^2}$. (b) Show that the average power falls to half its resonant value when $|\omega L - 1/\omega C| = R$. For a sharp resonance, deduce that the bandwidth is $\Delta\omega \approx R/L$. (c) A radio tuner has $L = 2.0 \mu\text{H}$, $R = 5.0 \Omega$ and $f_0 = 100 \text{ MHz}$. Find Q and the bandwidth, and compare the bandwidth with the 200 kHz spacing of FM channels.

P21.14

Displacement current. A parallel-plate capacitor with circular plates of radius R is being charged by a current I . (a) Show that the displacement current between the plates, $\epsilon_0 d\Phi_E/dt$, equals I . (b) Use the Ampère–Maxwell law to find the magnetic field between the plates at a distance $r < R$ from the axis.

P21.15

(a) For a plane wave with $\vec{E} = E_0 \sin(kx - \omega t) \hat{j}$ and $\vec{B} = B_0 \sin(kx - \omega t) \hat{k}$, apply Faraday’s law to a small rectangle in the xy -plane and show that $E_0 = (\omega/k)B_0 = cB_0$. (b) Show that the electric and magnetic energy densities of the wave are equal at every instant.

P21.16

Solar sail. A perfectly reflecting sail at 1 AU from the Sun faces the sunlight (1360 W/m^2). The sail and its payload have a mass of 10 g per square metre of sail. (a) Find the radiation pressure and the acceleration. (b) Compare this with the Sun's gravitational acceleration at 1 AU ($5.93 \times 10^{-3} \text{ m/s}^2$). (c) What is the largest mass per unit area at which the sail could “hover” against the Sun's gravity? Explain why this ratio does not depend on the distance from the Sun.

P21.17

RF heating in MRI. Safety rules limit the whole-body specific absorption rate (SAR) to 2.0 W/kg . (a) How much RF power may a 70 kg patient absorb? (b) Ignoring all cooling, how much would their body temperature rise in a 10-minute scan? Take the body's specific heat as 3500 J/(kg K) . (c) Why is *local* heating near metallic implants, or around loops formed by the patient's limbs or by ECG leads, a greater concern than the whole-body average?

P21.18

A mains notch filter. (a) Design a series LC branch, connected across a signal line, that resonates at 50 Hz using a $10 \mu\text{F}$ capacitor. Find the required inductance. (b) Comment on the practicality of such a component. Explain why modern ECG machines instead use active filters and digital signal processing, and why good electrode contact and a “driven right-leg” circuit are the first lines of defence against hum.

All worked examples and practice problems in these notes are original. Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

© 2026 Pratik Bhanuse. All rights reserved. · pratikbhanuse.github.io